

Draft - expert review pending

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Topic 2.2 - Water pollution and industrial effluents

A real classroom situation

In a settlement on the edge of Hyderabad, a hand pump stands twenty paces from an open drain. The drain carries grey water from houses and, twice a week, a dark discharge from a small textile unit upstream. A grandmother tells the health visitor, "The pump water looks clear, but three of my grandchildren have had loose motions this month." The visitor takes a sample bottle from her bag. Clear, she says, is not the same as clean.

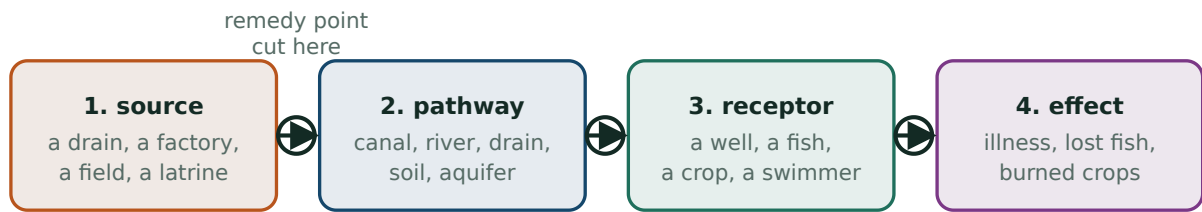
That drain and that pump are the whole of this topic in one picture: a source, a pathway, a receptor, and a health effect waiting to be measured.

Explanation

Water pollution: sources, types, impacts and remedies

Water pollution is the addition of substances or heat to water at a level that harms its use by people or by living things. Its logic is always the same chain: a **source** releases a pollutant, a **pathway** carries it, water in a river, canal, drain or aquifer, and a **receptor** receives it: a drinking well, a fish, a swimmer, a crop. Every remedy in this topic is a cut made at one link of that chain.

The pollution chain, and where to cut it



At the source

latrines and sewerage; keep waste out of the drain; effluent treatment before discharge; safer chemicals

On the pathway

protect the wellhead; lined drains; treatment of the water body itself where standards demand it

At the receptor

boil or chlorinate drinking water; a sealed, tested well; avoid irrigating crops with unsafe water

The rule: every remedy in this unit cuts one link.

Name the link before you choose the remedy.

Worked example: the Hyderabad drain is the source, the canal the pathway, the hand pump the receptor.

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The main pollutant types, with their usual sources:

- **Pathogens**, disease-causing organisms from human and animal waste. The world's heaviest water burden: the WHO estimates about 505,000 diarrhoeal deaths each year from microbiologically contaminated drinking water (WHO, 2023).
- **Organic matter**, food, slaughterhouse and dairy waste, that decays in water and uses up its oxygen, suffocating fish.
- **Nutrients**, nitrogen and phosphorus from fertiliser run-off and detergent, which over-fertilise water. The algal bloom that follows, and the oxygen collapse when it dies, is eutrophication.
- **Heavy metals**, chromium, lead, arsenic, from industry and old geology; they persist and accumulate.
- **Heat and suspended solids**, less discussed but real: hot discharge and silt both change what a water body can support.

Water pollutant types: source, impact, remedy

Pollutant type	Usual source	Impact in one line	One remedy
Pathogens	human and animal waste	diarrhoeal disease, the heaviest water burden	latrines, sewerage, safe water treatment
Organic matter	food, dairy, slaughterhouse	decays and uses up the water's oxygen; fish die	secondary (biological) treatment
Nutrients	fertiliser run-off, detergent	algal bloom, then oxygen buffer strips, phosphate collapse: eutrophication	limits, tertiary treatment
Heavy metals	industry: tanneries, plating	persist and accumulate in water, fish and people	tertiary treatment built for metals; NEQS limits
Heat and silt	hot discharge, erosion	change what the water body can support	cooling ponds; erosion control on banks

Clear is a colour, not a test: the deadliest pollutants in this table are all invisible.

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Remedies sit at the three links. At the **source**: latrines and sewerage, phasing phosphorus detergents, keeping waste out of the drain. At the **pathway**: treatment (chemical dosing, filtration), protecting wellheads, and the WHO's sanitation figures show what source control is worth: 1.4 million deaths a year are attributable to inadequate water, sanitation and hygiene (WHO, 2024). At the **receptor**: boiling or chlorinating drinking water, and closed wells with proper sealing.

Industrial effluents: sources, impacts, treatment and recycling

An effluent is the liquid waste a facility discharges. Textile dyeing, tanning, sugar, paper and steel each produce a characteristic one: tanneries carry chromium, dye houses carry colour and salts, sugar mills carry a heavy organic load. Untreated, the effects arrive down the chain: a drain that kills fish, land that burns crops irrigated with it, and groundwater that carries the metals for decades.

Treatment runs in stages, each removing a different kind of load.

- **Primary treatment** screens and settles: grit and a share of solids drop out. It is cheap and removes the visible load.
- **Secondary treatment** puts bacteria to work on the dissolved organic matter, in tanks or trickling filters, and settles their remains. This is the step that restores oxygen to the receiving water.
- **Tertiary treatment** polishes what is left: filtration, adsorption and special steps for metals, salts or nutrients, so the water can be reused or discharged safely.

Recycling closes the loop. A factory that treats its own effluent can return it to washing, cooling or irrigation instead of the river, which cuts both its draw on fresh water and its discharge. Pakistan's National Environmental Quality Standards, the discharge limits the law sets, are the yardstick a treatment plant is built to meet; the standards name the pollutant, the receiving water and the maximum concentration allowed.

A worked example. Take the Hyderabad drain. The textile unit is the source; primary treatment would settle its fibre bits, secondary would eat its dye organics, tertiary would treat its salts and colour. The household grey water needs a different, cheaper chain, and the well needs a sealed head and a test. One drain, three links, different remedies at each.

Common misconception. "Water that looks clear is safe." The most dangerous pollutants are invisible: pathogens by the million, dissolved metals, nitrate. Clear is a colour, not a test. The only clear water you may trust is water that was treated or boiled after it came from a protected source.

Activity: Trace the drain

Work in pairs. Your teacher gives each pair a sketch map of a neighbourhood with one canal, six buildings (houses, a workshop, a clinic) and two hand pumps marked. Mark every possible source of water pollution on the map, draw the pathway arrows to the canal and the pumps, and mark who or what receives each. Then choose the single cheapest remedy that would cut the most arrows. Allow 20 minutes, then compare maps with another pair.

Check your understanding

1. Name the three links of the pollution chain and give one example of each from your own surroundings. (*Remember*)
2. Which pollutant type causes eutrophication, and what happens in the water when it does? (*Understand*)
3. A tannery discharges chromium. Which treatment stage is built for it, and what remains if that stage is skipped? (*Apply*)
4. Explain why "the water looks clear" is not evidence that a well is safe. Use two different pollutant types in your answer. (*Analyze*)

Summary

Water pollution runs source, pathway, receptor, effect, and every remedy cuts one of those links. The pollutant types, pathogens, organic matter, nutrients, heavy metals, heat and silt, each pair a usual source with a matching impact and remedy. Industrial effluents carry a concentrated mix and are treated in stages, primary for solids, secondary for organics, tertiary for metals, salts and colour, after which the water can be recycled instead of discharged. Clear water is not tested water: the deadliest pollutants are invisible.

Self-assessment checklist

Sign in to sync your ticks across devices.

- ☐ I can name the three links of the pollution chain and a remedy at each link.
- ☐ I can match each of the five pollutant types to its usual source and impact.
- ☐ I can explain the three treatment stages and what each removes.
- ☐ I can explain why water recycling serves both the factory and the river.

Try this at your practicum school

Find the school's nearest drain or watercourse. Walk ten minutes along it, note what enters it (pipes, open channels, dumped waste) and where it goes. Draw

the chain, source, pathway, receptor, for the two biggest inputs, and name one remedy for each that a neighbourhood committee could actually ask for. Do not name the school or any person.

Summative task

Write 170 to 200 words on the following.

Your neighbourhood committee has one budget line for water. Using the pollution chain and the treatment stages, argue for the single best use of that money, and name one thing that should NOT be funded first, with reasons for both.

Mini-rubric

Criterion	What a strong response does
Chain used (Analyze)	Argues from source-pathway-receptor, not from a wish list
Stages understood (Understand)	Matches the treatment stage to the pollutant actually present
Trade-off named (Evaluate)	Says no to something plausible and explains why it can wait
Clarity and register	Plain sentences for a committee; within the word count

Further reading

- World Health Organization. (2023). *Drinking-water* (fact sheet): <https://www.who.int/news-room/fact-sheets/detail/drinking-water>
- World Health Organization. (2024). *Sanitation* (fact sheet): <https://www.who.int/news-room/fact-sheets/detail/sanitation>

- Park, J. E., & Park, K. *Park's Textbook of Preventive and Social Medicine*. (guide's closest real work, cited by reference only; flagged per D-2026-0001)

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